



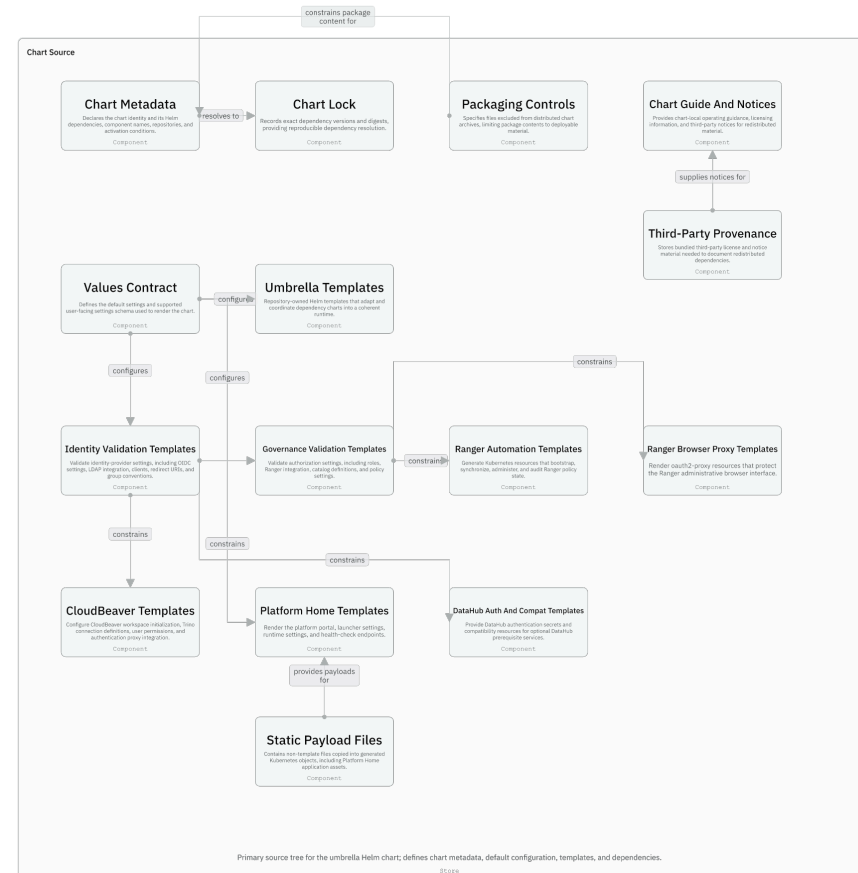
Session 04 / How the chart works

Chart Internals, Values Contract, And Render Flow

One hour handover deck

40-45 min narrative / 10-15 min checkpoint /
5 min Q&A

The chart is a shared contract plus umbrella-owned glue around upstream dependencies.



Session path

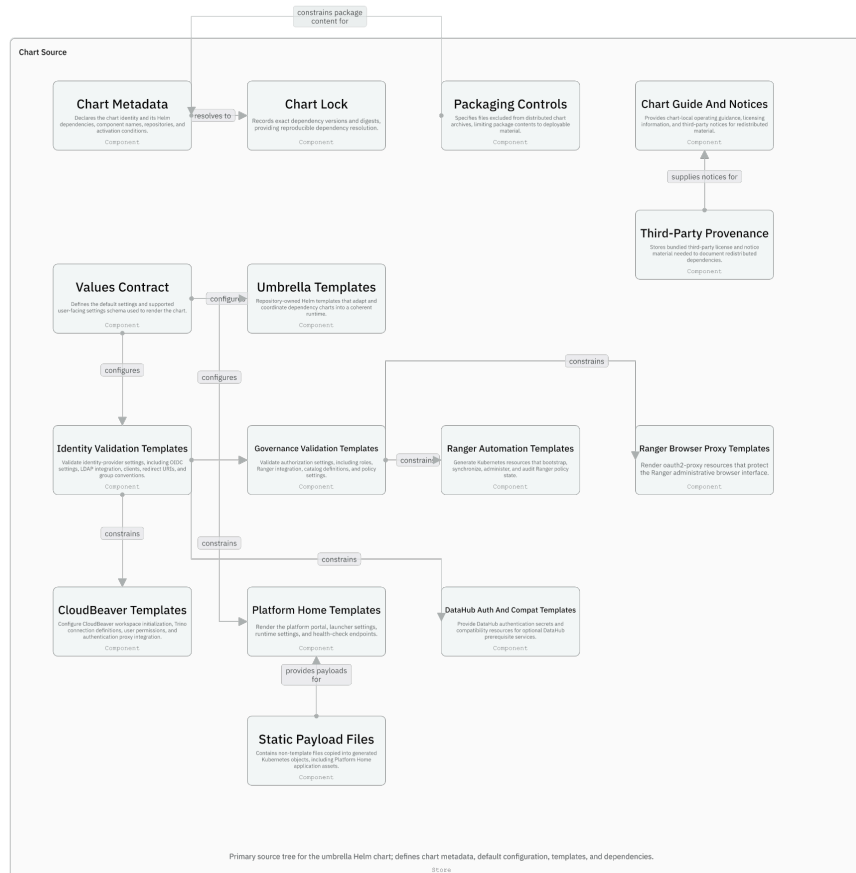
- 1 **Ownership layers**
- 2 **Values categories**
- 3 **Validation templates**
- 4 **Lookup behavior**

Session rhythm

40-45 min narrative
10-15 min checkpoint
5 min Q&A

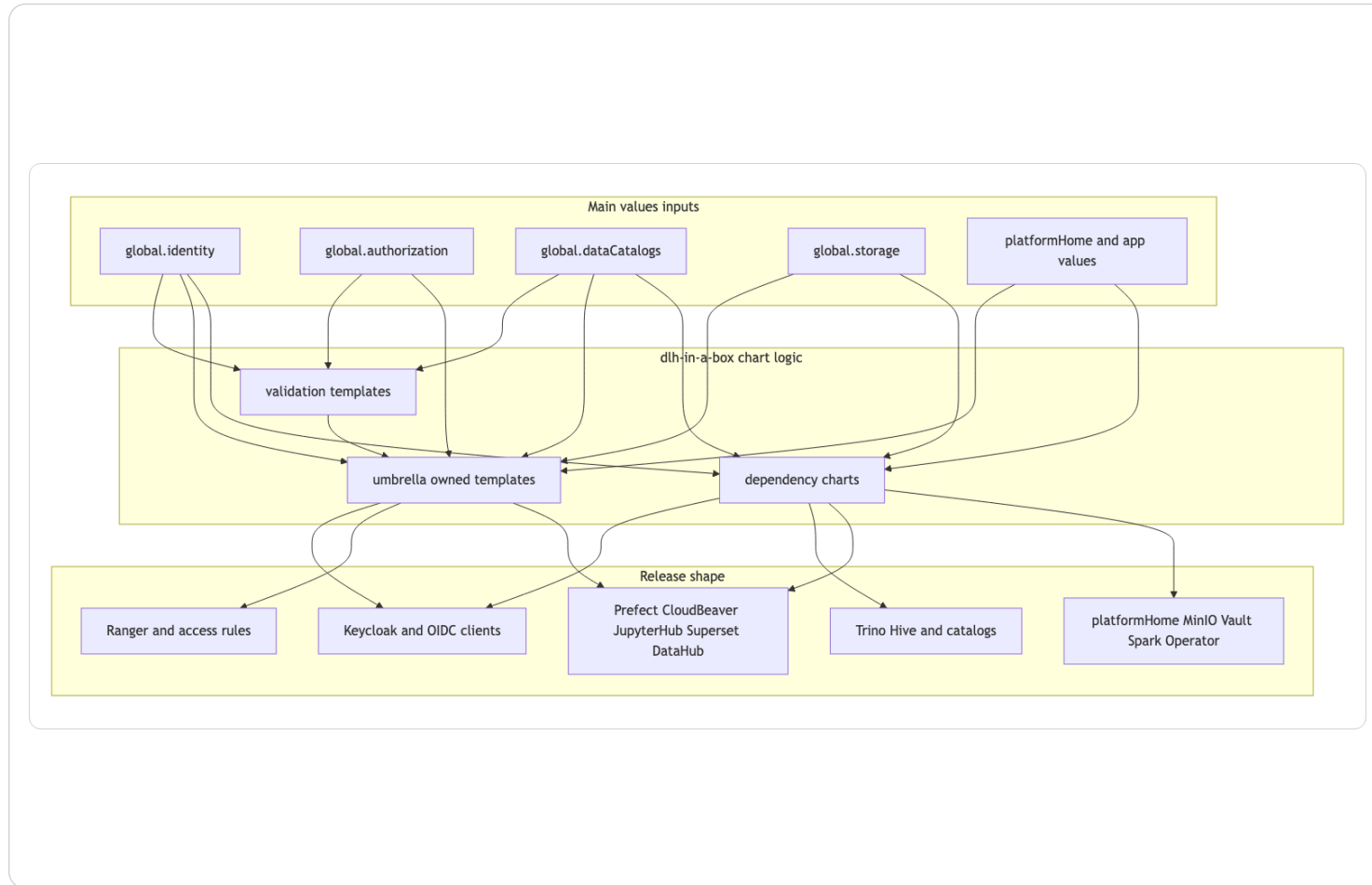
Presenter notes carry the script and demo prompts.

Chart source components



- Chart.yaml and Chart.lock define packaged dependency state.
- values.yaml and values.schema.json define the public input surface.
- templates/ contains umbrella-owned behavior.
- third_party/ and notices travel with packaged material.

Values drive validation, templates, and dependencies



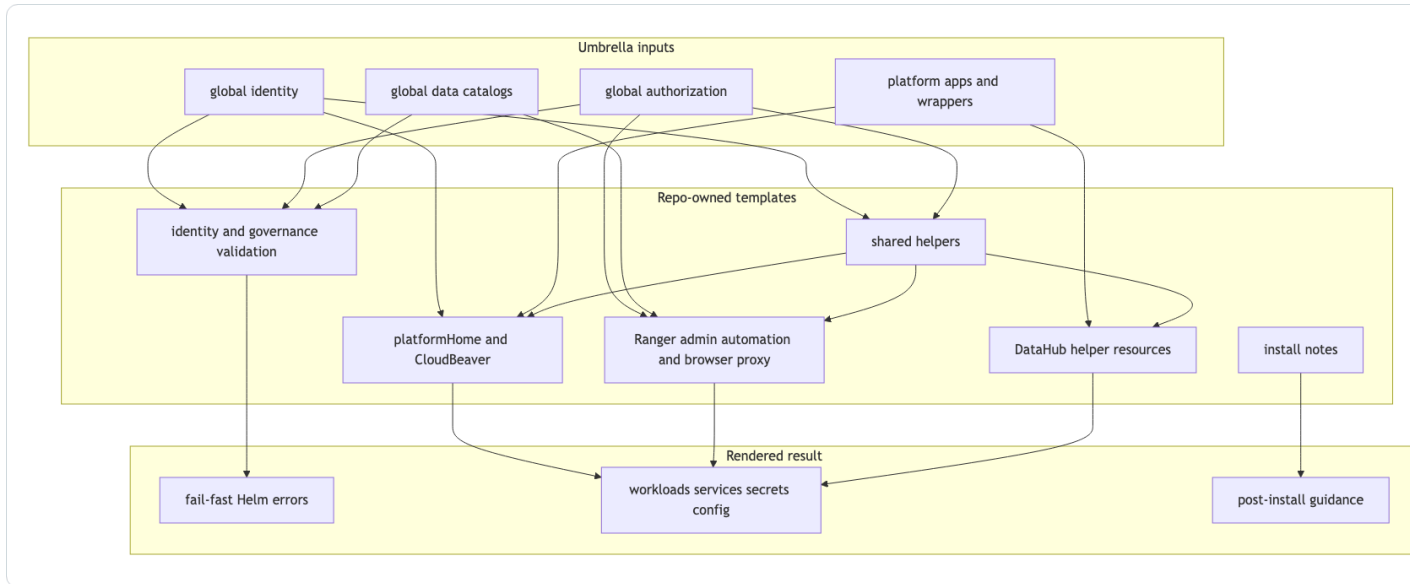
- `global.identity`, `global.authorization`, `global.storage`, and `global.dataCatalogs` are cross-component contracts.
- App-specific sections configure repo-owned wrappers.
- Dependency sections pass through upstream chart values.

Read values.yaml by category

Category	Examples	What changes
Shared contract	global.identity, global.authorization, global.storage, global.dataCatalogs	Cross-component behavior
Repo-owned apps	platformHome, cloudbeaver, auth proxy sections	Umbrella-specific logic
Dependency pass-through	keycloak, minio, superset, jupyterhub, datahub	Upstream chart configuration
Local subcharts	hive, hivePostgresql	First-party metastore behavior

Umbrella templates coordinate the runtime

- Validation templates fail fast on unsupported combinations.
- platformHome, CloudBeaver, Ranger, and DataHub helpers adapt upstream services.
- Template behavior is often where product assumptions live.



Fail-fast validation lives in more than schema

File	What it blocks
values.schema.json	Input shape and allowed structural forms
identity-validation.yaml	Unsupported identity modes, missing client wiring, app auth inconsistencies
governance-validation.yaml	Broken platform roles, incomplete governance metadata, unsafe authorization settings

Helm lookup makes upgrade renders different

Upgrade-sensitive behavior

- CloudBeaver can read existing Secrets for rollout checksums.
- DataHub auth material is preserved across upgrades.
- DataHub prerequisite compatibility can mirror existing MySQL Secrets.
- Trino helpers can read existing S3 credentials for generated catalogs.

**CloudBeaver
can read
existing
Secrets for
rollout
checksums.**

Guided checkpoint: trace a values rule

```
Open charts/dlh-in-a-box/values.yaml
```

```
Open charts/dlh-in-a-box/values.schema.json
```

```
Open charts/dlh-in-a-box/templates/identity-  
validation.yaml
```

```
Render one positive and one negative contract  
fixture with ./hack/render-contract.sh
```

Checkpoint questions

- Can they explain whether a rule belongs in schema or a validation template?
- Can they explain what a render-contract fixture is proving?

Behavior-heavy files worth reading first

Area	Start file
platformHome launchpad and helper API	templates/platform-home.yaml
Ranger automation and audits	templates/ranger-automation.yaml
CloudBeaver bootstrap and trust behavior	templates/cloudbeaver.yaml
DataHub auth and prerequisite compatibility	templates/datahub-auth-secrets.yaml; datahub-prerequisites-compat.yaml
Ranger browser boundary	templates/ranger-browser-proxy.yaml

Session 04 landing

What the inheriting team should now be able to do

- They can read values by ownership category.
- They know schema is not the full validation story.
- They can explain why lookup affects upgrade behavior.
- They know the main behavior-heavy template entrypoints.

Leave-behind

Use the PPTX notes as a presenter script and source-map.md as the reference trail.